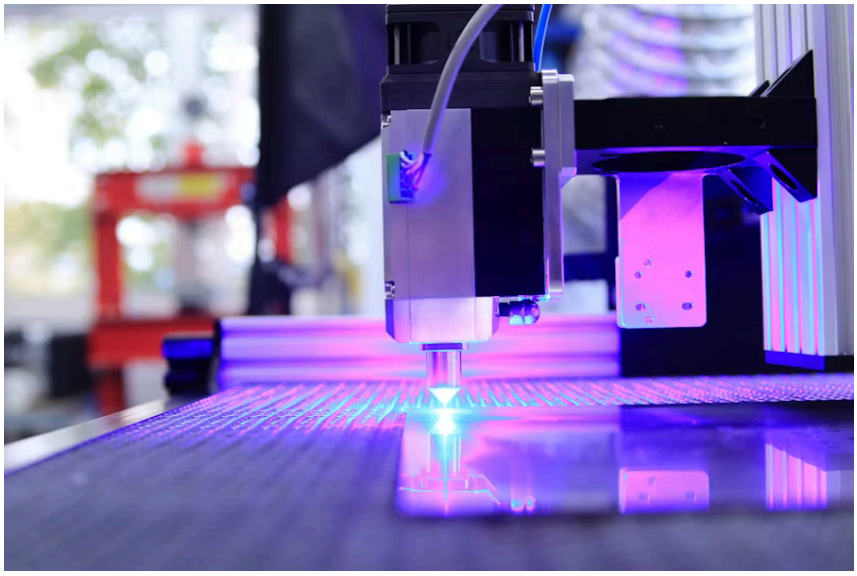
THE CLANDESTINE SIDE OF 3D PRINTING

By Zain Hamdia

Beyond Prototypes and Plastic



3D printing is paving the way in advancing the engineering world. Although 3D printing is a very useful tool, many don't know of its capabilities and application in the everyday world.

3D printing can help with building robots, prototype products of all kinds, parts for cars and even the cars themselves using a special type of durable, heat resistant filament! More recently and quite amazingly, clay and cement 3d printers have been used to build entire homes. The advances in 3D printing have been truly remarkable.

The Secret Technology Hidden in Plain Sight

While these uses alone have been innovative, the industry has a little known secret that borders on covert technology. It is called PolyVinyl Alcohol or PVA filament. Essentially it is a dissolvable filament. Interesting in concept, but more importantly, could solve a huge problem in today's consumer-driven world.

The Shipping Crisis

It all begins with the shipping industry, who are desperately trying to keep up with the demands of consumers to receive items at home. While this has been going on for years, the impact of COVID has forced us into a new level of shipping needs. The pandemic has increased online shopping by 60%!

Behind all of this shopping is packing and boxes. Lots and lots of boxes. When you receive a package, it is typically in a cardboard box. 80% of items sold are shipped or packed in a cardboard box. The US alone uses more than 80 billion boxes per year.

The Environmental Cost of Cardboard

Another sobering issue to consider is the number of trees needed in this process. 1 dead tree is equal to about 150 cardboard boxes and we, as a nation, use 100 billion cardboard boxes every year. What?? Yes. That means we kill around 666,666,667 trees every year for our cardboard boxes. Are you concerned yet?

Each person that receives a package has 2 options: recycle it or throw it in the garbage. Approximately 70% of the population recycles their cardboard boxes. One might think recycling is a good option, but it doesn't actually reduce the amount of the cardboard in circulation, it simply turns it into reusable cardboard.

Meanwhile, companies are still making brand new boxes to meet demands. This will create an overflow of boxes and the only way to truly rid ourselves of them is by throwing them away which, in turn, harms the environment. When cardboard is at a garbage dump, it gets burned and when this happens it releases mass amounts of methane. Methane is known to contribute to 20% of global warming. Or it is left to decompose, which can take five years or more.

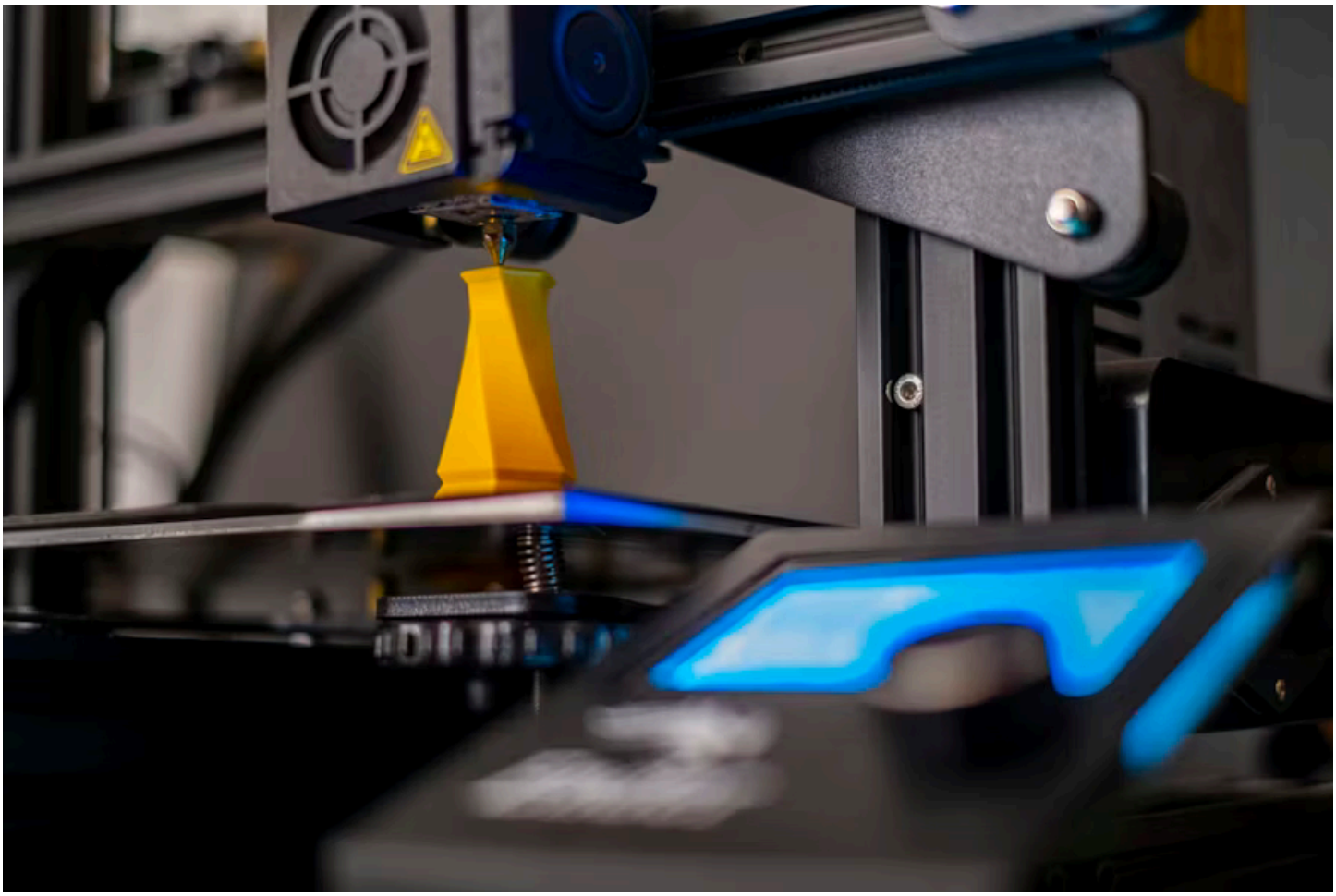
The Hidden Problems with Recycling

The next best option that isn't tossing your boxes in a landfill is recycling. Recycling plants do have their downsides though. Not only are they unsafe for the environment, they are unsafe for their workers. The employees of these plants are exposed to toxic fumes let off by the machines and inhale large amounts of dust composed of wood, paper and tiny little bits of metal. At the end of all of this, the structural quality of recycled products is poor compared to the new product.

A Dissolvable Solution

There is an environmentally safe solution to all of this. PVA filament. PVA is very special because it dissolves in water without leaving any toxic or environmentally harmful result. This makes it possible to 3d print the packaging for our products, have them shipped to us and get rid of the waste created by the packaging in a clean way.

So what if it rains while your package is in route or once it is delivered? Not a problem. It takes 12-18 hours for the packaging to fully dissolve the material. If you don't want to go through that, recycling is still an option. The leftover package can be ground up and formed into new filament which would be printed as new packages.



In concept, the solution is simple, but the process to get there is not. If we were to make this transition, shipping companies would need many of these printers running simultaneously, the printers themselves would need to function at a much faster speed and the PVA filament would need to be mass produced. None of this is impossible but will take time and initial cos

We may get there or we may not, but you have to admit - making your boxes disappear sounds pretty good.